FUNDAMENTALS OF CYBER SECURITY
MODULE 1 — HANDWRITTEN QUESTIONS & ANSWERS

Exam-ready notes prepared from the 8 questions listed in Module 1 of the uploaded FCSDF Question Bank.

Q1. Explain classification of cybercrime with the help of examples.

Answer:

Cybercrime is an illegal activity in which a computer, digital device, network or the internet is used as a tool, target or place of crime.

Classification of cybercrime:

1) Crimes against individuals

- Identity theft: Stealing personal information and using it without permission.
- Phishing: Sending fake emails or messages to steal passwords or bank details.
- Cyberstalking: Repeatedly harassing or threatening a person online.
- Online defamation: Publishing false information to damage someone's reputation.

2) Crimes against property

- Data theft: Copying confidential files without authorization.
- Intellectual property theft: Illegally copying software, music or designs.
- Ransomware: Encrypting a victim's files and demanding money to restore them.
- Cybersquatting: Registering a domain name similar to a famous brand for misuse.

3) Crimes against organizations

- Hacking: Gaining unauthorized access to an organization's systems.
- DoS/DDoS attack: Flooding a server with traffic so genuine users cannot access it.
- Insider attack: An employee misuses authorized access to steal or damage data.
- Malware attack: Using malicious software to disrupt operations or

steal information.

4) Crimes against society or government

- Cyberterrorism: Using digital attacks to create fear or disrupt essential services.
- Spreading fake news: Deliberately circulating false information online.
- Website defacement: Changing a public or government website without permission.
- Cyber espionage: Secretly collecting confidential government or defense information.

Example: An attacker sends a fake bank link and steals a customer's login details. This is phishing and may lead to identity and financial fraud.

Conclusion: Cybercrime affects individuals, property, organizations and governments. Awareness and suitable security controls help prevent it.

Q2. Analyze the various factors that contribute to the creation of vulnerabilities in cyber systems.

Answer:

A vulnerability is a weakness in hardware, software, network design or security procedures that an attacker may exploit.

Factors that create vulnerabilities:

1. Software bugs: Programming errors and insecure code may allow unauthorized access.
2. Outdated software: Old operating systems and applications may contain known weaknesses.
3. Weak passwords: Short, common or reused passwords make accounts easier to compromise.
4. Misconfiguration: Default passwords, open ports and incorrect permissions expose systems.
5. Human error: Clicking suspicious links or sharing credentials can create security gaps.
6. Poor access control: Giving users unnecessary permissions increases possible damage.
7. Insecure networks: Unencrypted Wi-Fi and unsafe remote access

may expose information.

8. Lack of security updates: Delayed patching leaves known vulnerabilities uncorrected.

9. Third-party risks: External vendors, libraries or services may introduce weaknesses.

10. Physical security failures: Stolen devices or unlocked server rooms can expose data.

Example: A company uses an old web application with an unpatched SQL injection weakness. An attacker may exploit it to access or modify the database.

Preventive measures:

- Apply security patches regularly.
- Use strong passwords and multi-factor authentication.
- Perform vulnerability assessments and penetration testing.
- Follow secure coding and least-privilege practices.
- Train employees and maintain backups.

Conclusion: Vulnerabilities arise from technical, human and organizational weaknesses. Continuous assessment and timely correction reduce the attack surface.

Q3. Explain Types of Cybercriminals.

Answer:

Cybercriminals are individuals or groups who use computers, networks or the internet to perform illegal activities.

Types of cybercriminals:

1. Black-hat hackers: Break into systems without permission for theft, damage or personal gain.

2. White-hat hackers: Authorized security professionals who test systems and report weaknesses. They are ethical hackers when working with permission.

3. Grey-hat hackers: Access systems without authorization, often without malicious intent, but their actions may still be illegal.

4. Script kiddies: Inexperienced attackers who use ready-made tools or scripts developed by others.

5. Organized cybercrime groups: Professional groups involved in

ransomware, financial fraud and data theft for profit.

6. Insiders: Employees, contractors or former employees who misuse legitimate access intentionally or accidentally.

7. Hacktivists: Attack systems to promote a political, social or ideological cause.

8. State-sponsored attackers: Groups supported by a nation-state for espionage or strategic cyber operations.

9. Cyber terrorists: Use cyberattacks to create fear, disrupt essential services or threaten national security.

10. Phishers and scammers: Use fake websites, emails, calls or messages to steal money or confidential information.

Example: A ransomware group attacks a hospital, encrypts patient records and demands payment. This is an example of an organized cybercrime group.

Conclusion: Cybercriminals differ in skill, motivation and resources. Understanding their types helps organizations choose suitable security measures.

Q4. Analyse the concepts of vulnerabilities, threats, and risks in the context of information security.

Answer:

Vulnerability, threat and risk are related concepts in information security, but each has a different meaning.

1. Vulnerability:

A vulnerability is a weakness in a system that can be exploited.

Examples: weak password, outdated software, unpatched server or incorrect firewall settings.

2. Threat:

A threat is any possible event, person or action that can exploit a vulnerability and cause harm.

Examples: hacker, malware, phishing email, insider misuse, fire or hardware failure.

3. Risk:

Risk is the possibility of loss or damage when a threat exploits a vulnerability. It depends on the likelihood of an incident and the

impact it may cause.

Relationship:

Threat + Vulnerability □ Exploitation □ Impact or Loss

Risk is assessed by considering the likelihood and severity of the impact.

Example:

- Vulnerability: A company's server has an unpatched weakness.
- Threat: An attacker attempts to exploit that weakness.
- Risk: Customer data may be stolen, causing financial loss and reputational damage.

Risk management steps:

1. Identify important assets and vulnerabilities.
2. Identify possible threats.
3. Estimate likelihood and impact.
4. Prioritize risks as high, medium or low.
5. Apply controls such as patching, encryption and access control.
6. Monitor and review risks regularly.

Conclusion: A vulnerability is a weakness, a threat is a possible cause of harm, and risk is the potential loss resulting from their interaction.

Q5. Evaluate the importance of the CIA Triad in cybersecurity.

Answer:

The CIA Triad is a basic information security model consisting of Confidentiality, Integrity and Availability. It helps organizations protect information and systems.

1. Confidentiality:

It ensures that information is accessible only to authorized users.

Methods: authentication, authorization, encryption and strong passwords.

Example: Only authorized bank employees can view customer account details.

2. Integrity:

It ensures that information remains accurate, complete and is not

changed without authorization.
Methods: hashing, digital signatures, access permissions and audit logs.
Example: A student's examination marks must not be changed by an unauthorized person.

3. Availability:
It ensures that information and services are available to authorized users whenever required.
Methods: regular backups, disaster recovery, redundant servers and protection against DoS attacks.
Example: An online banking service should remain accessible to customers.

Importance of the CIA Triad:
- Provides a foundation for security policies and controls.
- Protects sensitive and confidential information.
- Prevents unauthorized modification of data.
- Supports reliable access to systems and services.
- Helps assess risks and design safeguards.

Conclusion: All three principles are important. A system is not secure if data is confidential but inaccurate, or accurate but unavailable. Effective cybersecurity balances all three.

Q6. Write a short note on Internet Governance.

Answer:
Internet Governance is the development and application of shared principles, rules, standards and decision-making procedures that shape the use and development of the internet.

Objectives:
1. Maintain a stable, secure and reliable internet.
2. Support affordable and accessible internet for everyone.
3. Protect privacy and freedom of expression.
4. Encourage fair competition and innovation.
5. Coordinate technical standards and internet resources.
6. Address issues such as cybercrime, spam and misuse of data.

Major participants:

- Governments: Create laws and policies for internet use and cybersecurity.
- Private sector: Provides internet services, platforms, infrastructure and technology.
- Technical community: Develops and maintains internet protocols and standards.
- Civil society: Represents users and raises concerns about rights and access.

Conclusion: Internet Governance requires cooperation among governments, businesses, technical experts and users to keep the internet open, stable and secure.

Q7. Difference between Passive and Active attack.

Answer:

A passive attack attempts to obtain or monitor information without changing system resources or data. An active attack attempts to modify, damage, disrupt or inject information into a system.

Passive attack:

- Main aim: To secretly observe or collect information.
- Data: Usually does not modify the data.
- Detection: Difficult to detect because system operation may remain normal.
- Prevention: Encryption and secure communication.
- Examples: Eavesdropping, traffic analysis and packet sniffing.

Active attack:

- Main aim: To alter data, disrupt services or gain unauthorized control.
- Data: May modify, delete or insert data.
- Detection: Often easier to notice because system behaviour may change.
- Prevention: Authentication, access control, firewalls, monitoring and integrity checks.
- Examples: Denial of Service (DoS), masquerade, replay and message modification.

Key differences:

- Passive = monitoring; Active = modification or disruption.
- Passive attacks mainly affect confidentiality; active attacks may affect integrity and availability.
- Encryption helps protect against passive attacks, while active attacks need several security controls.

Conclusion: Both attacks threaten information security. Organizations should use encryption, authentication, monitoring and strong access controls to reduce them.

Q8. Describe the Need for a Comprehensive Cyber Security Policy.

Answer:

A Cyber Security Policy is a formal set of rules and guidelines that explains how an organization protects its computers, networks, applications and information.

Need for a comprehensive cyber security policy:

1. Protects information: Helps maintain confidentiality, integrity and availability of data.
2. Defines responsibilities: Clearly explains the security duties of employees, managers and IT teams.
3. Controls access: Specifies who can access systems and what actions they are allowed to perform.
4. Prevents cyberattacks: Establishes rules for passwords, updates, antivirus, firewalls and safe internet use.
5. Supports incident response: Defines how to identify, report, contain and recover from security incidents.
6. Ensures compliance: Helps the organization follow applicable laws, regulations and standards.
7. Reduces human error: Provides guidelines and training for safe use of email, devices and networks.
8. Protects business continuity: Encourages backups, recovery planning and continuity procedures.
9. Manages third-party risk: Sets security requirements for vendors and service providers.
10. Builds security awareness: Creates a culture in which everyone understands their role in cybersecurity.

Key components:

- Purpose and scope.
- Roles and responsibilities.
- Acceptable use of IT resources.
- Password and access-control rules.
- Data classification and protection.
- Network, device and remote-access security.
- Incident reporting and response.
- Backup, recovery and employee training.
- Policy review and enforcement.

Conclusion: A comprehensive cyber security policy provides a common security framework, reduces risks and helps an organization protect its digital assets.